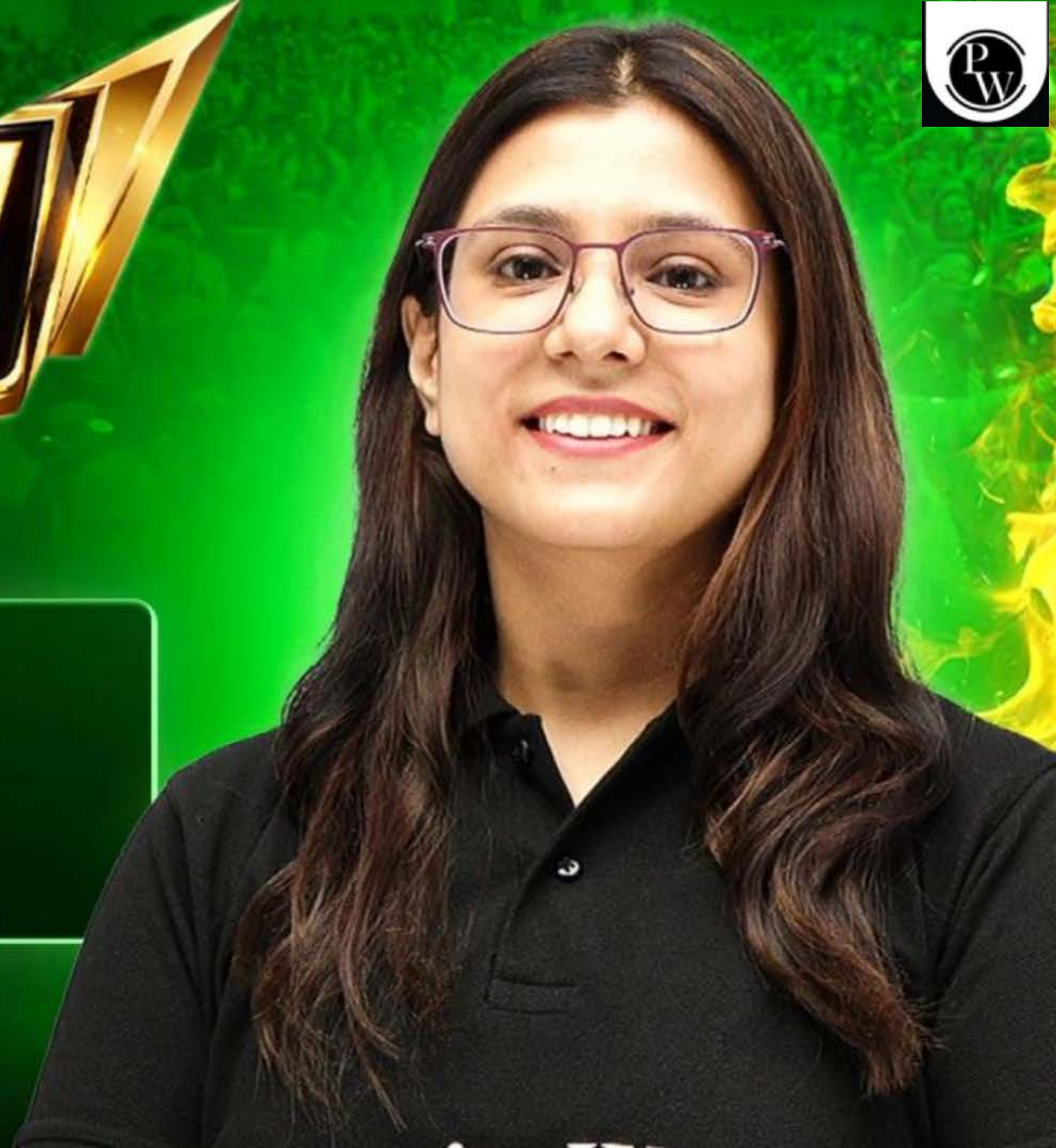
VICTORY 2027

Sense Organs

LECTURE-4 BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. Ear & its Function
2. Most imp Ques.




HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

EAR

The human ear is concerned with two functions, hearing and body balance.

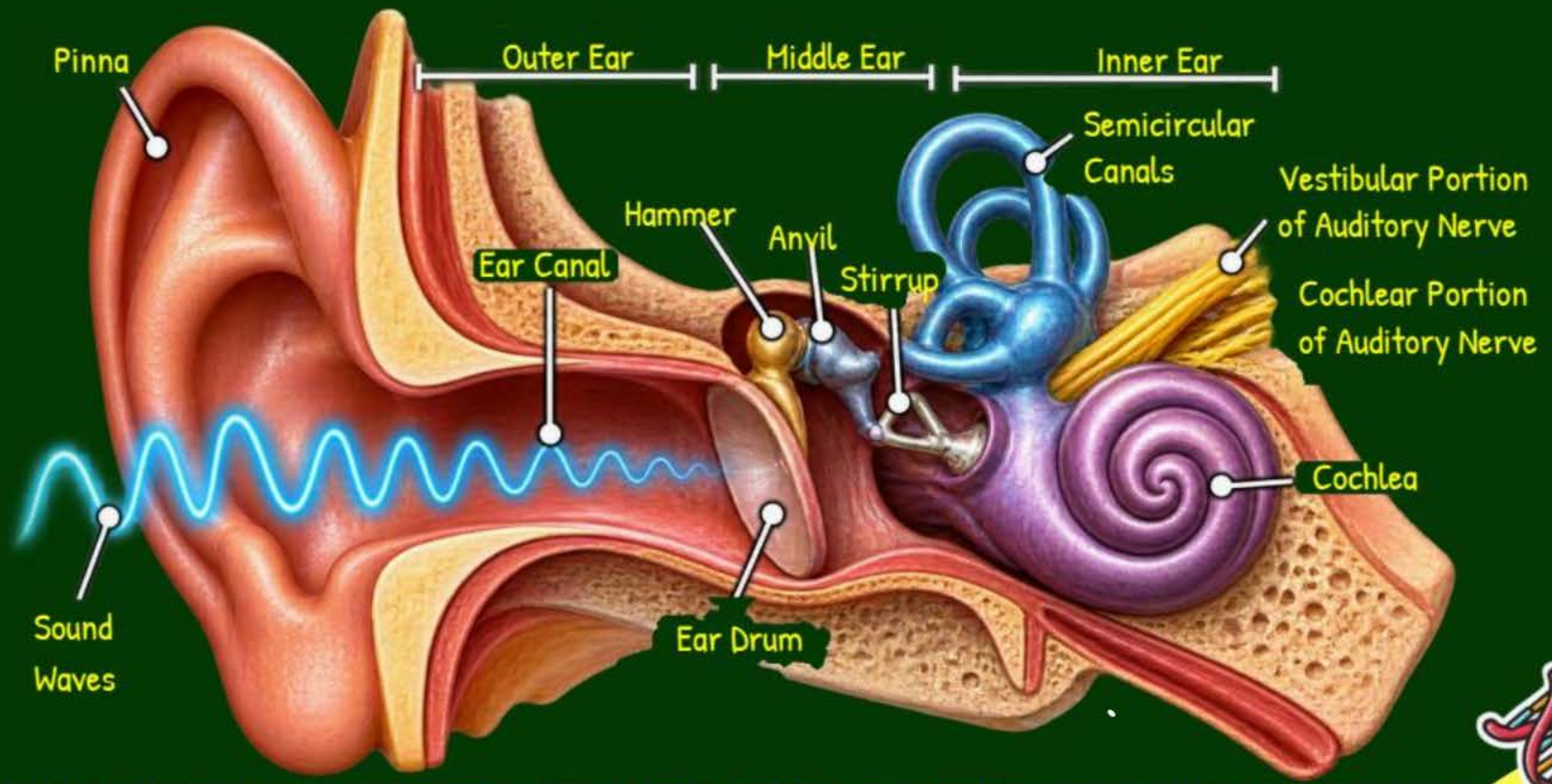
It has three main divisions :

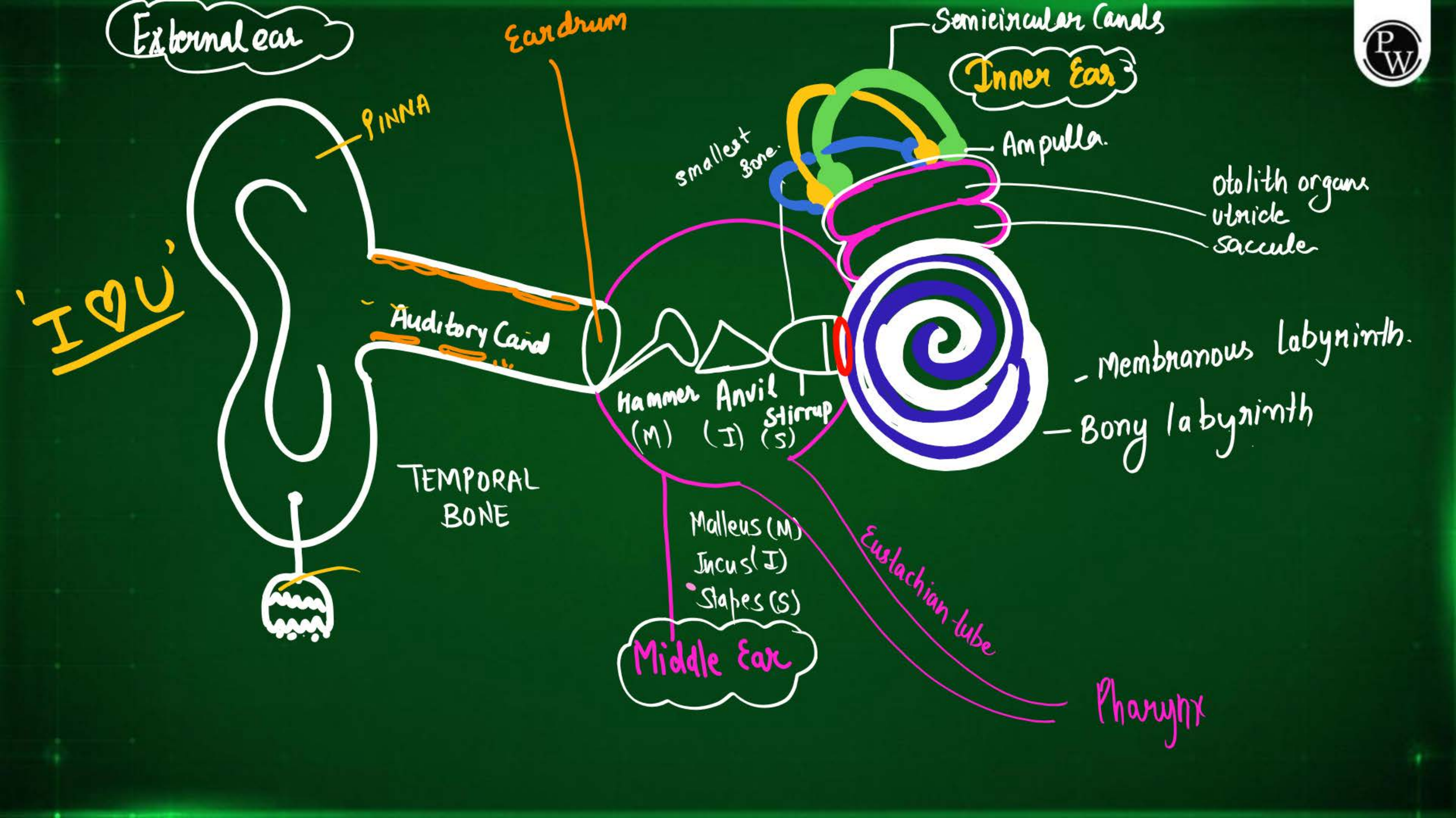
(i) Outer ear,

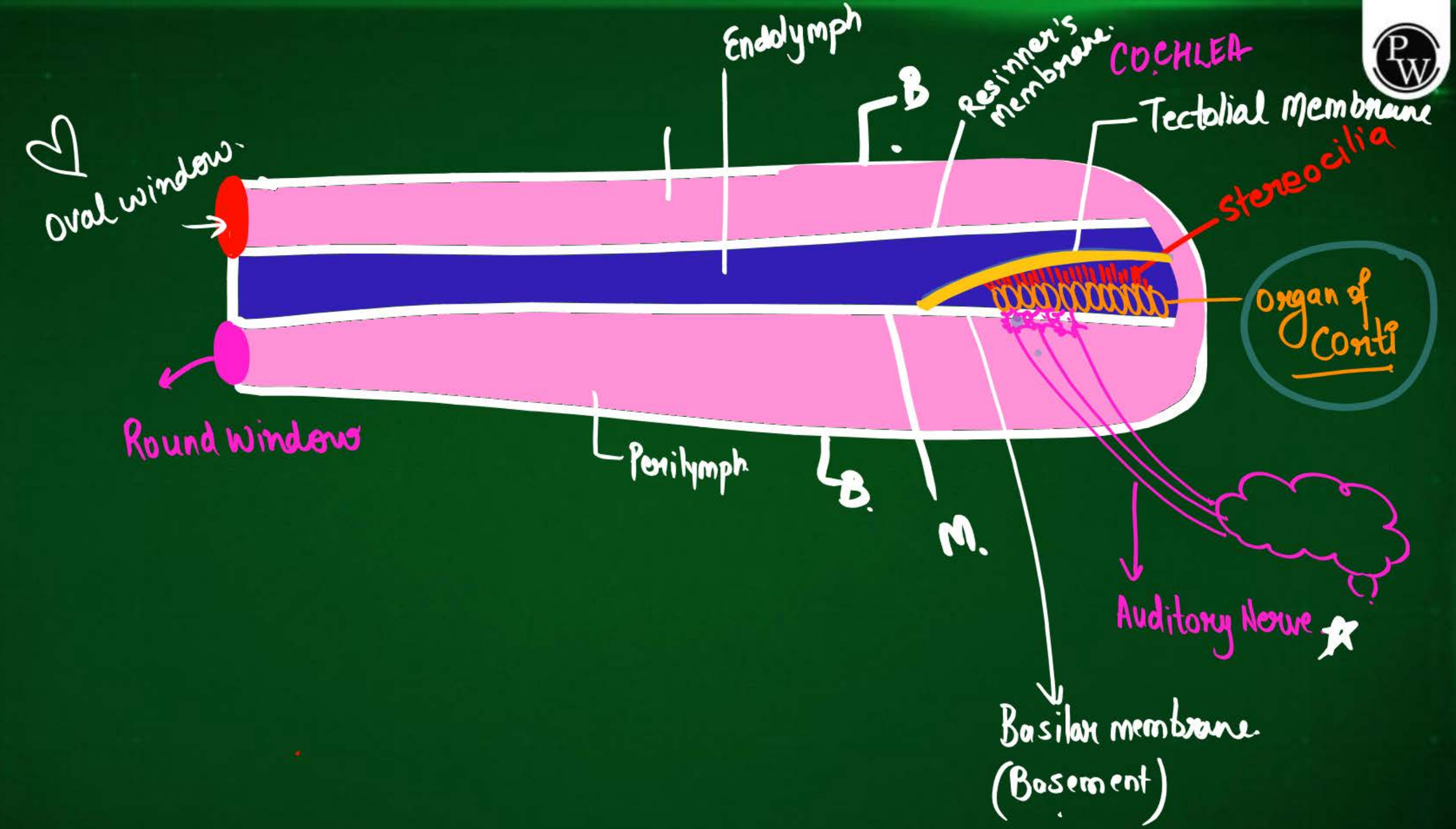
(ii) Middle ear and

(iii) Inner ear.

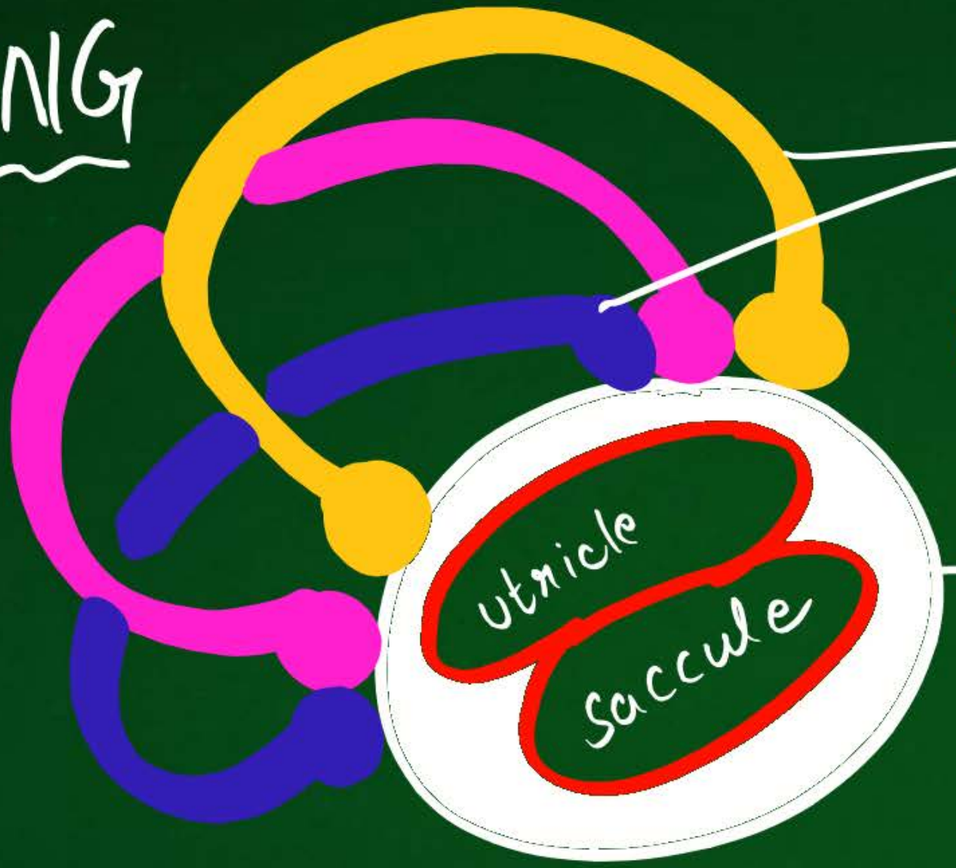
Vestibulo Cochlear Nerve




External ear
Eardrum
Semicircular Canals
Inner Ear
PINNA
smallest Bone
Ampulla
otolith organs
utricle
saccule
'I ♥ U'
Auditory Canal
Membranous labyrinth
Bony labyrinth
Hammer (M)
Anvil (I)
Stirrup (S)
TEMPORAL BONE
Malleus (M)
Incus (I)
Stapes (S)
Middle Ear
Eustachian tube
Pharynx


Endolymph
B
Reissner's membrane
COCHLEA
Tectorial Membrane
Oval window
Stereocilia
Organ of Corti
Round window
Perilymph
B
M
Auditory Nerve
Basilar membrane
(Basement)

BALANCING



3 semi circular canals

Dynamic Equilibrium

Otolith Organs

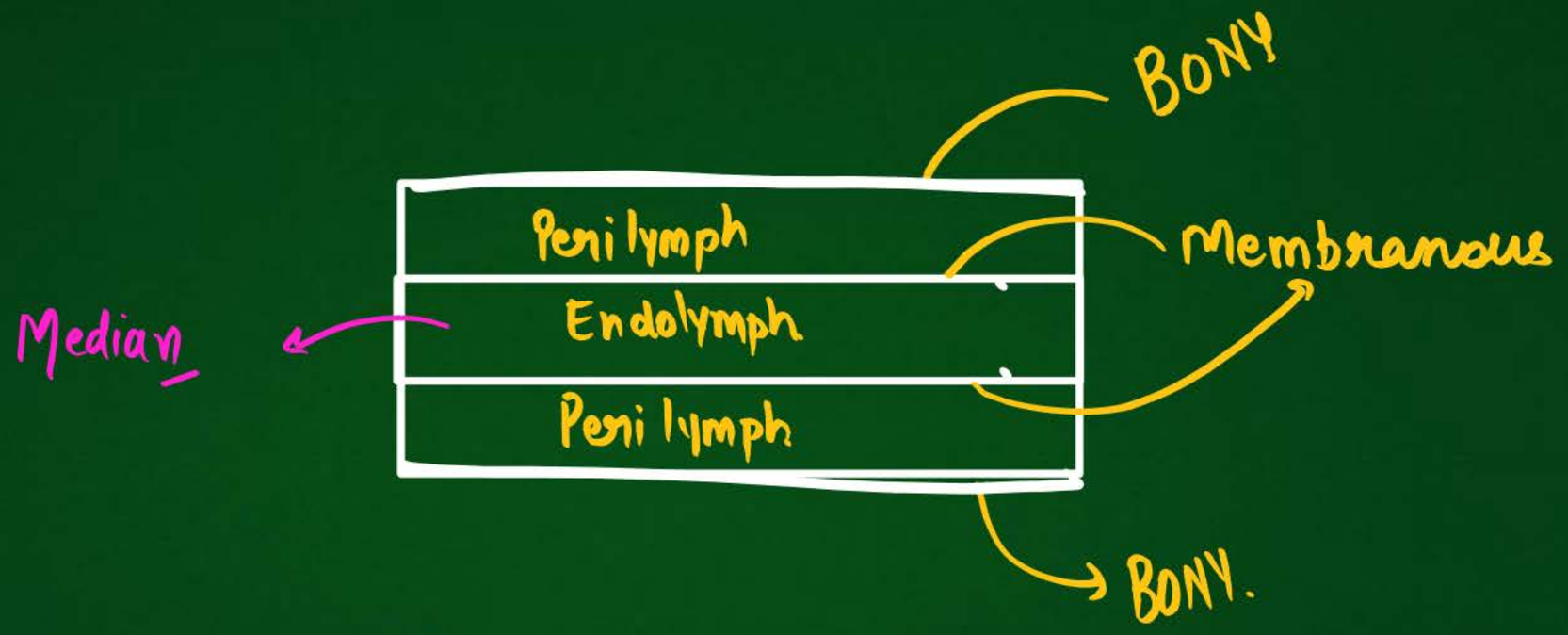
Vestibule.

Static Equilibrium

Bony & Membranous labyrinth present.

Perilymph & Endolymph present.

Vestibular Apparatus


BONY
Membranous
Median
Peri lymph
Endolymph
Peri lymph
BONY.

MAHA LIVE CLASS

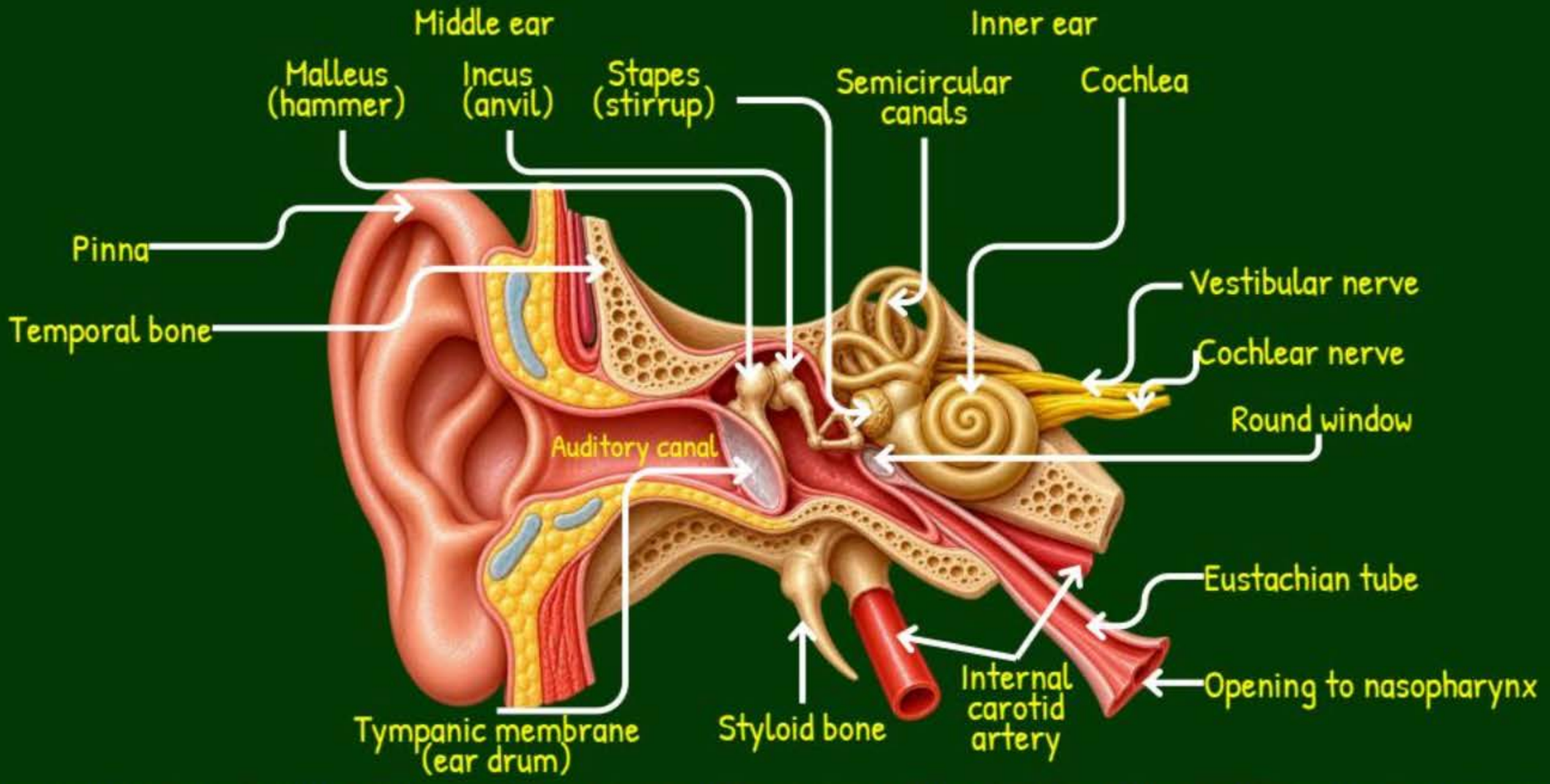
SUNDAY

5PM

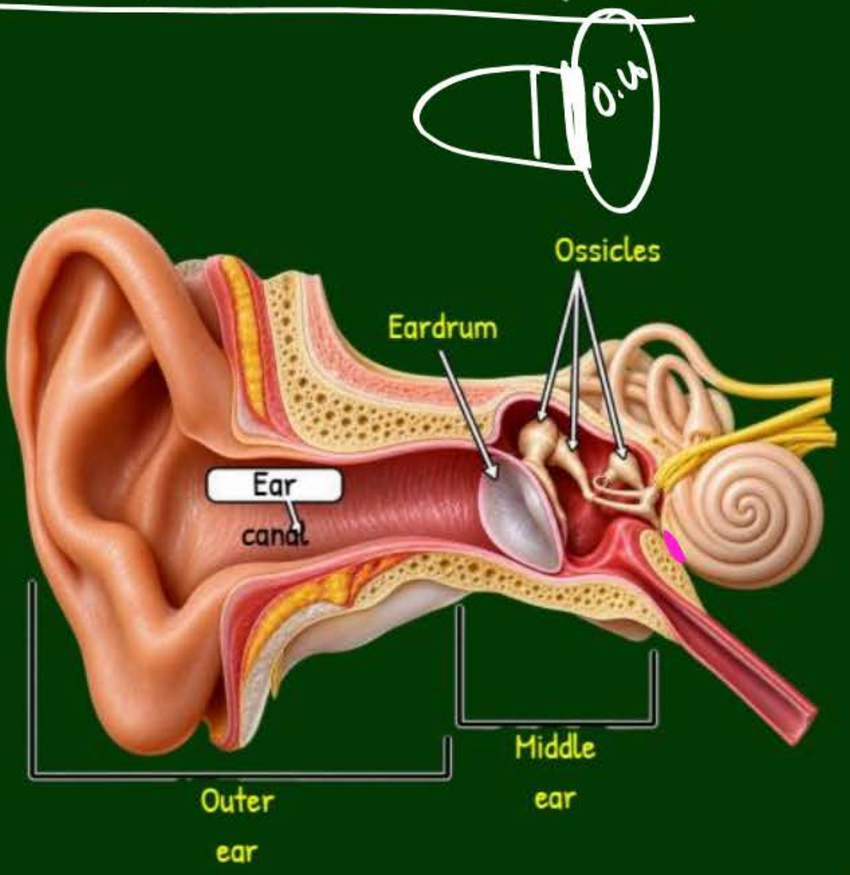
2 Hours Batch

EAR

(i) The outer ear consists of the projecting part pinna (also called "auricle") and the passage auditory canal leading to the ear drum (or tympanum).

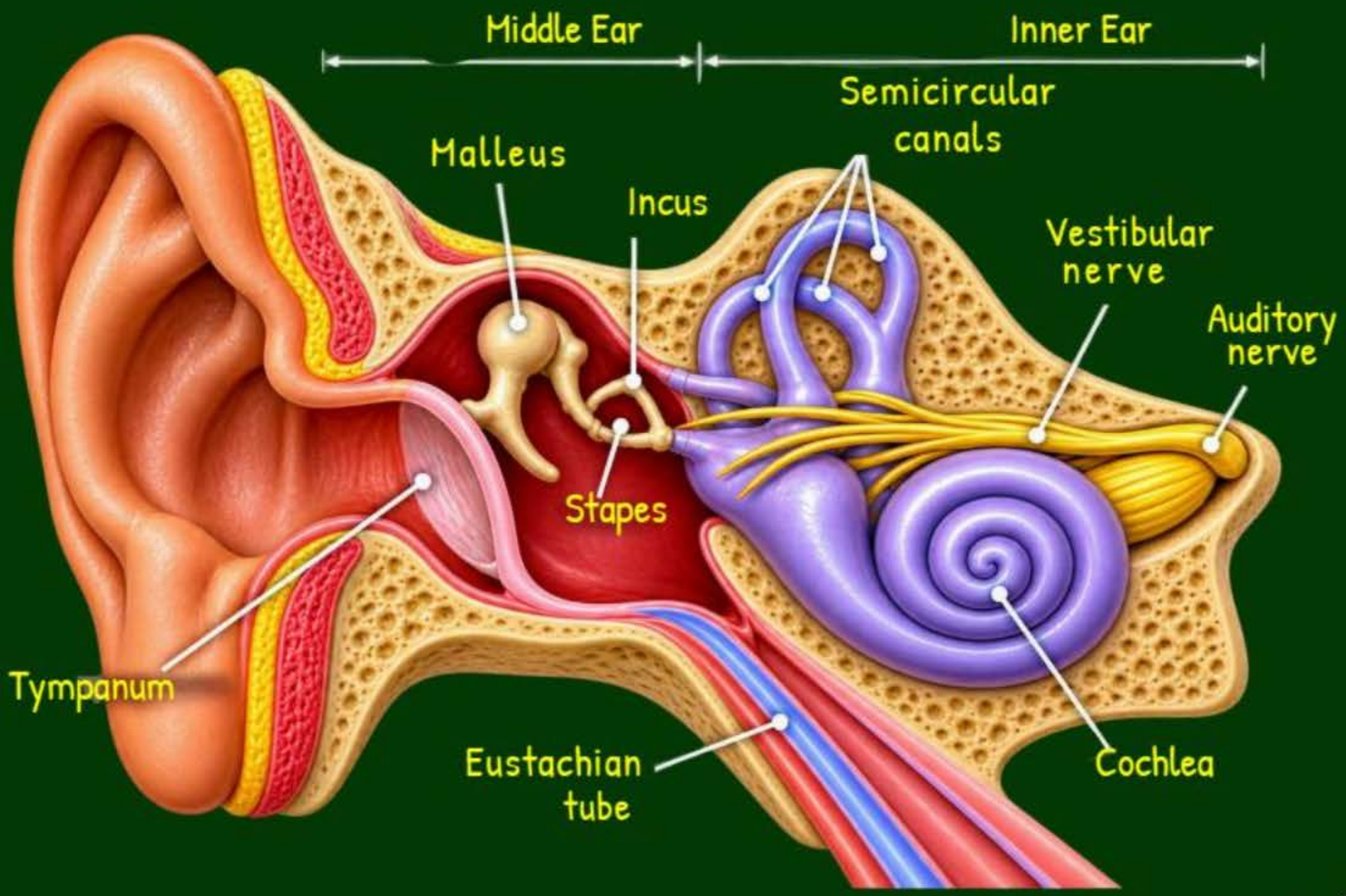


- Ear ossicles
- The middle ear contains three tiny bones malleus, incus and stapes or hammer, Anvil, Stirrup
- The handle of the hammer bone is attached to the inner surface of the ear drum.
- Its opposite end is connected to the anvil which, in turn, is joined to the stirrup.
- Oval window -A membrane-covered opening leading to the inner ear .The flat part of the stirrup fits on it.
- Round window-A second opening,covered by a thin membrane, connects the middle and the inner ear.



EAR

(iii) The inner ear or membranous labyrinth has three parts - the cochlea, semicircular canals and the vestibule. The cochlea is spiral-shaped and looks like a snail shell.



EAR

- It has two and a half turns. Its inner winding cavity is divided into three parallel canals separated by membranes.



- The median (cochlear) canal second is filled with a fluid called endolymph and the other two (1 and 3) with perilymph.

- The middle canal contains areas possessing sensory cells, spiral organ called organ of Corti for hearing. The nerve fibres arising from these cells join the auditory nerve. The sensory cells lie on the basilar membrane.



EAR

Semi-circular canals - arranged at right angles to each other in three different planes so that one is horizontal and the other two are vertical. One end of each canal is widened to form an ampulla which contains sensory cells for dynamic balance while the body is in motion and nerve fibres from them join the auditory nerve.

Vestibule –it has two parts

- a utriculus and a sacculus, joined at the base of semicircular canal. These parts also contain sensory cells for static balance when the body is stationary as in standing.

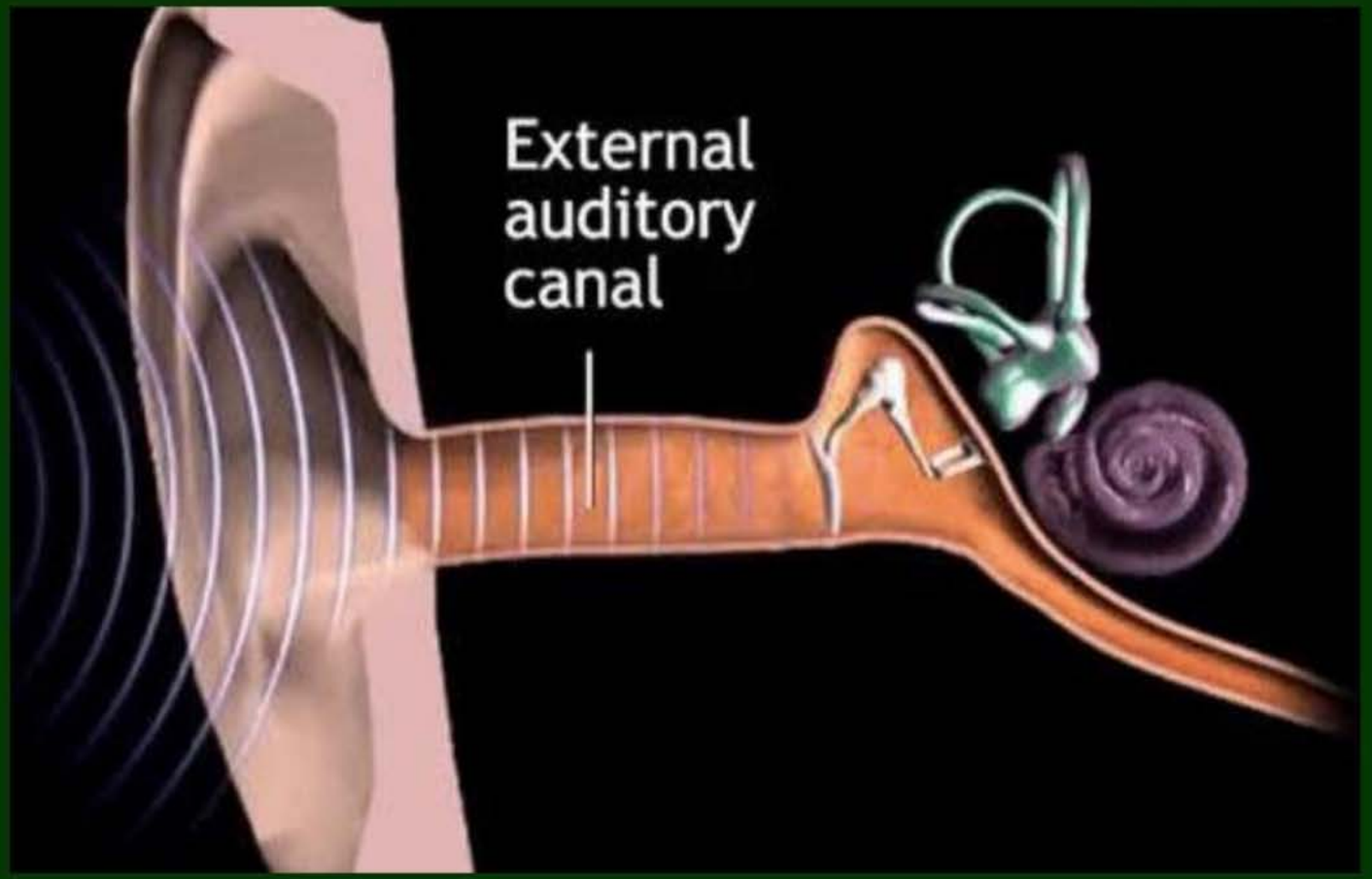


FUNCTIONS OF THE EAR

- The internal ear is involved in two sensory functions: hearing and body balance.
- The pinna collects the sound waves and conducts them through the external auditory canal. They finally strike on the ear drum which is set into vibration.
- The eustachian tube equalises the air pressure on either side of the ear drum allowing it to vibrate freely.

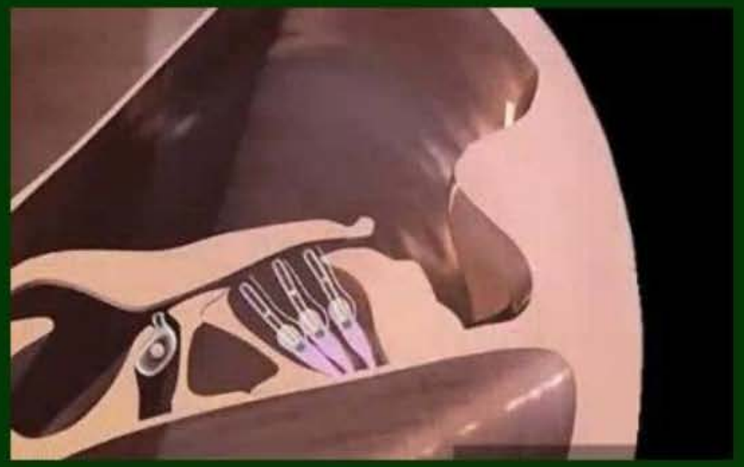





External
auditory
canal

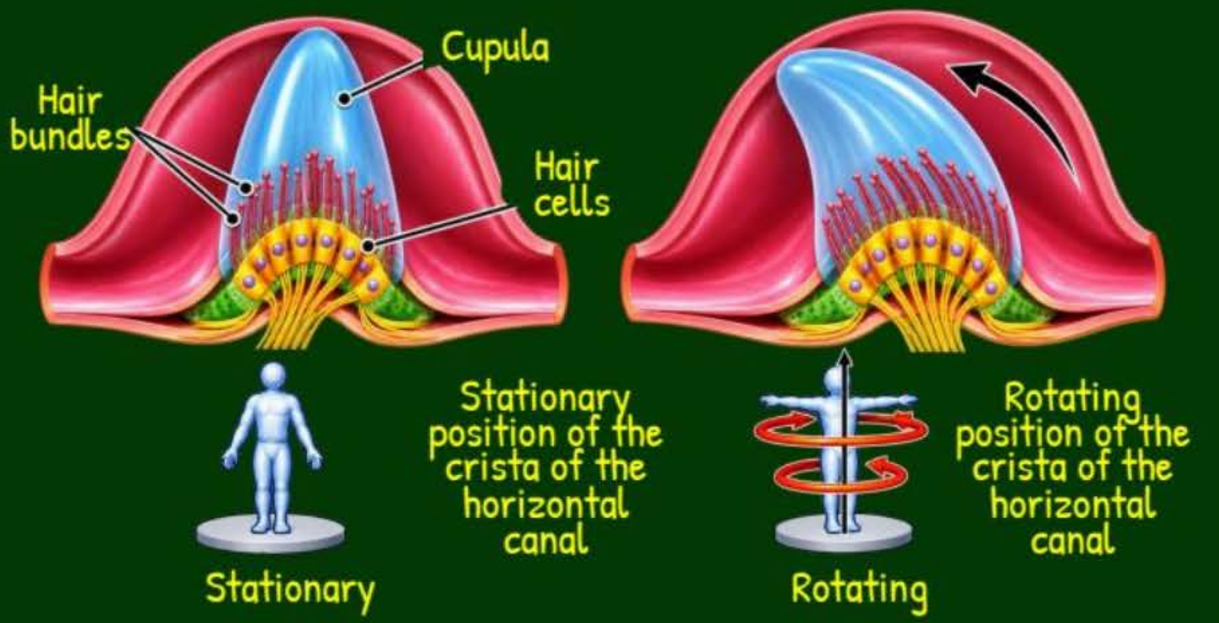
EAR

- The vibrating ear drum sends vibrations to three ossicles.
- The ear ossicles amplify the sound.
- The vibrating stirrup transmits the vibration to the oval window which in turn vibrates the perilymph.
- The vibrating movements of the endolymph stimulate the hair-like processes of the sensory cells of the cochlea (in spiral organ)
- The impulses are transmitted to the brain via the auditory nerve.



BALANCING

- **Static Equilibrium (Static Balance)**• It helps you maintain balance when your body is at rest or moving in a straight line (linear motion).
- It is maintained by the vestibule, which consists of two parts: the utriculus and the sacculus.•



STATIC EQUILIBRIUM – EXAMPLES (DAILY LIFE)


Standing upright


Sitting on a chair


Lying down


Tilting head to one side


Bending head forward or backward


In a lift moving upward (acceleration)


In a lift moving downward (deceleration)


Car accelerating forward


Car braking (deceleration)


Travelling in a train (straight line)


Standing on an inclined surface


Walking on a rope (maintaining position)


Standing on a gymnastic beam/rod (maintaining position)


Standing on a ladder


Standing with a school bag


Holding a heavy object


Pushing a wall (maintaining position)


Leaning on a table


Standing on one leg


Standing on tiptoes


Standing on an uneven surface


Sitting on the floor


Lying on stomach


Floating in water (maintaining position)


Standing in a bus/train (maintaining position)


Maintaining a yoga posture


Sleeping (maintaining position)


Foetus in uterus (maintaining position)

DYNAMIC EQUILIBRIUM – EXAMPLES (DAILY LIFE)


Turning head left and right


Looking up and down


Rotating head (spinning)


Spinning while dancing


Cartwheel


Somersault


Rotating on ice (skating)


Rotating on a chair


Merry-go-round


Taking a turn while cycling


Turning a car


Turning a bike


Turning while dancing (spin)


Swinging a bat


Swinging a racquet


Throwing a ball


Jumping and rotating in air


Diving (rotation)


Skiing (turning)


Skateboarding (turning)


Changing direction while running


Turning while swimming


Twisting in air (gymnastics)

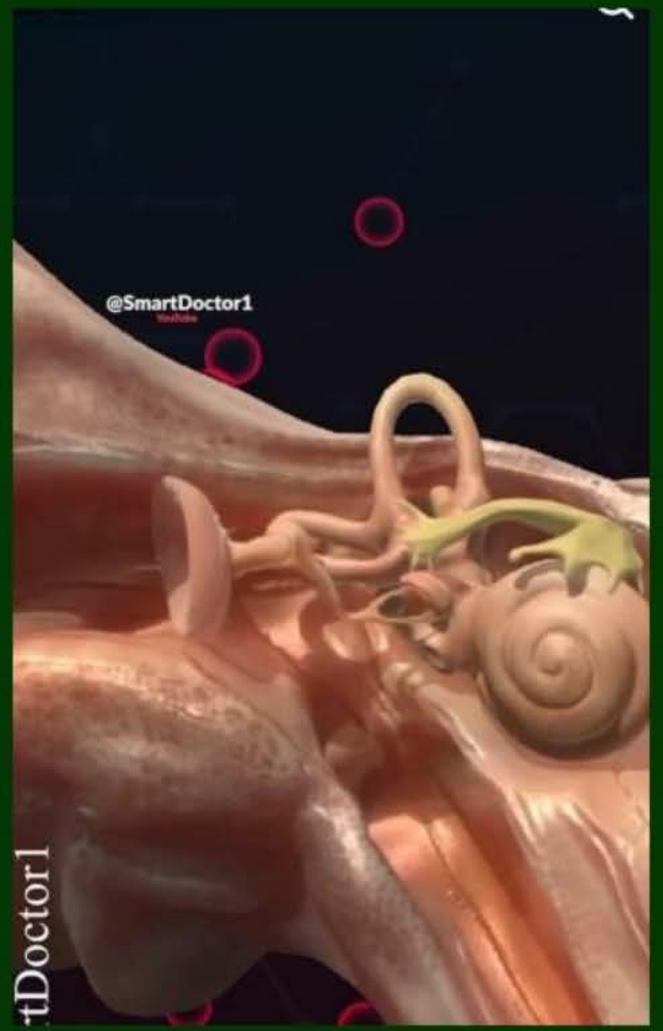

Hula hooping

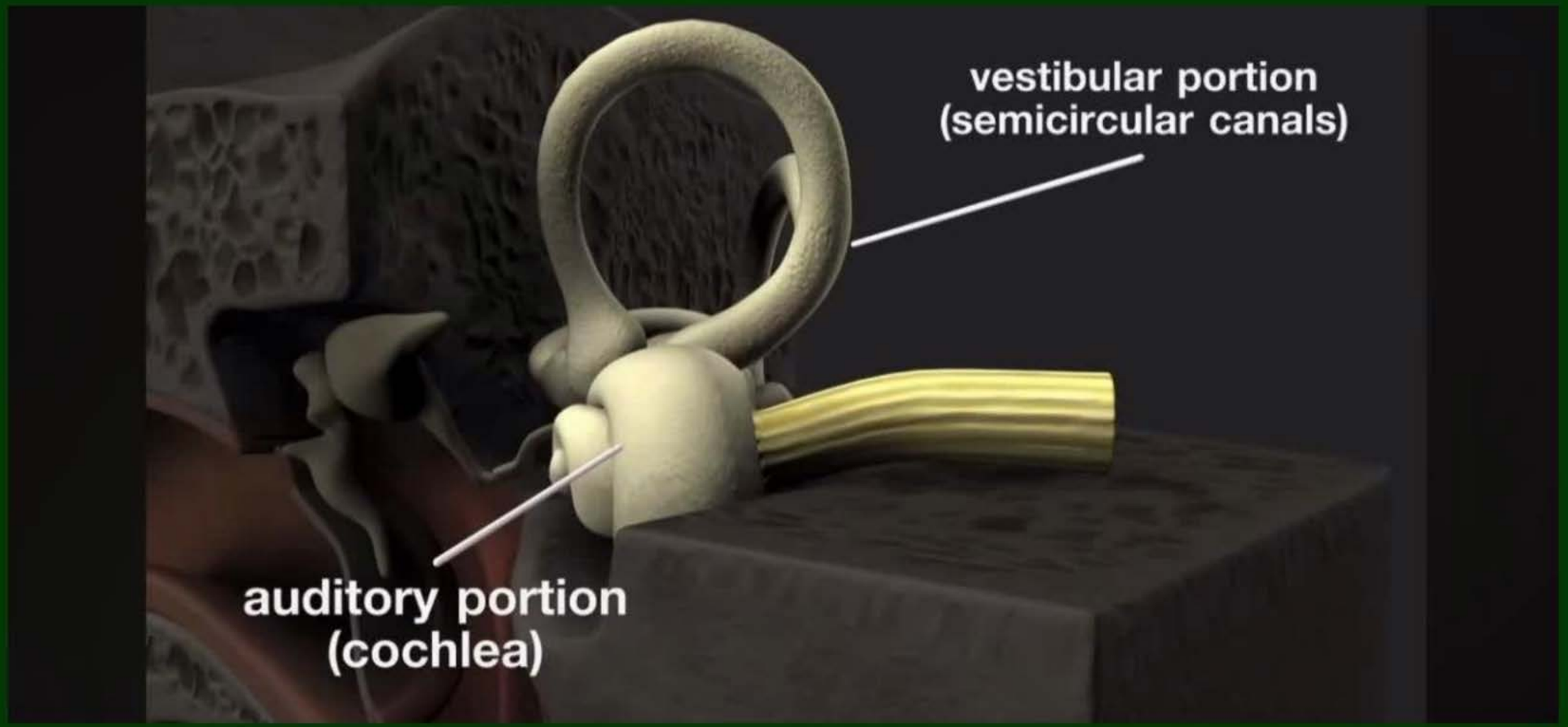


BALANCING

- **Dynamic Equilibrium (Dynamic Balance)•**
- It helps you maintain balance when your body is in rotational motion (spinning, turning, or twisting).
- It is maintained by the three semicircular canals.
- These canals are arranged at right angles to each other and are filled with a fluid called endolymph.




@SmartDoctor1
tDoctor1


vestibular portion
(semicircular canals)
auditory portion
(cochlea)

(iv) Tanya, a ten-year-old girl had a severe earache. Her mother took her to an E.N.T. doctor for a checkup. The doctor noticed that she had a throat infection which had spread to her ear. Answer the following questions:

(a) Name the structure that connects the throat and ear.

(b) What is the collective term for the three cartilaginous structures present in the middle ear? MIS Ear Ossicles

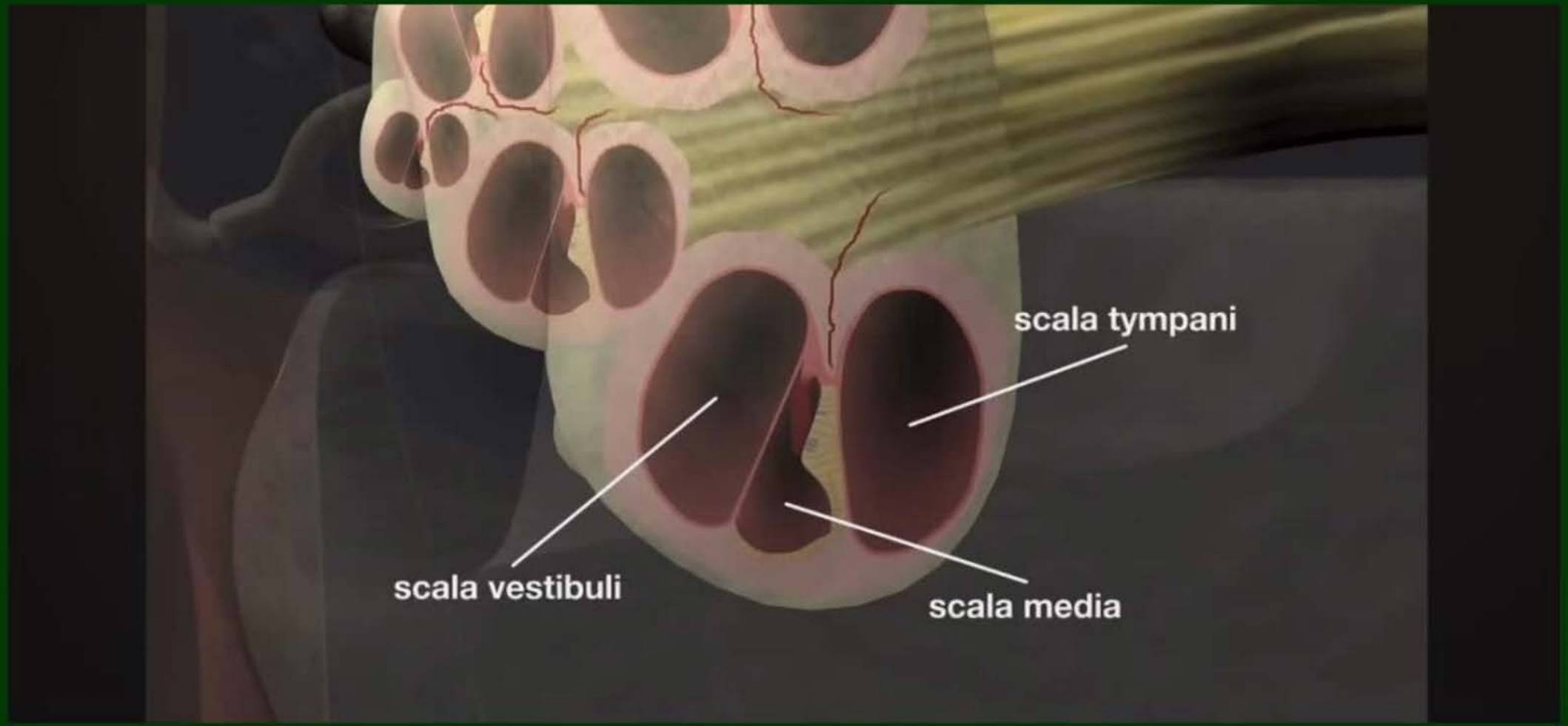
(c) Which is the sensory organ located in cochlea? Organ of Corti

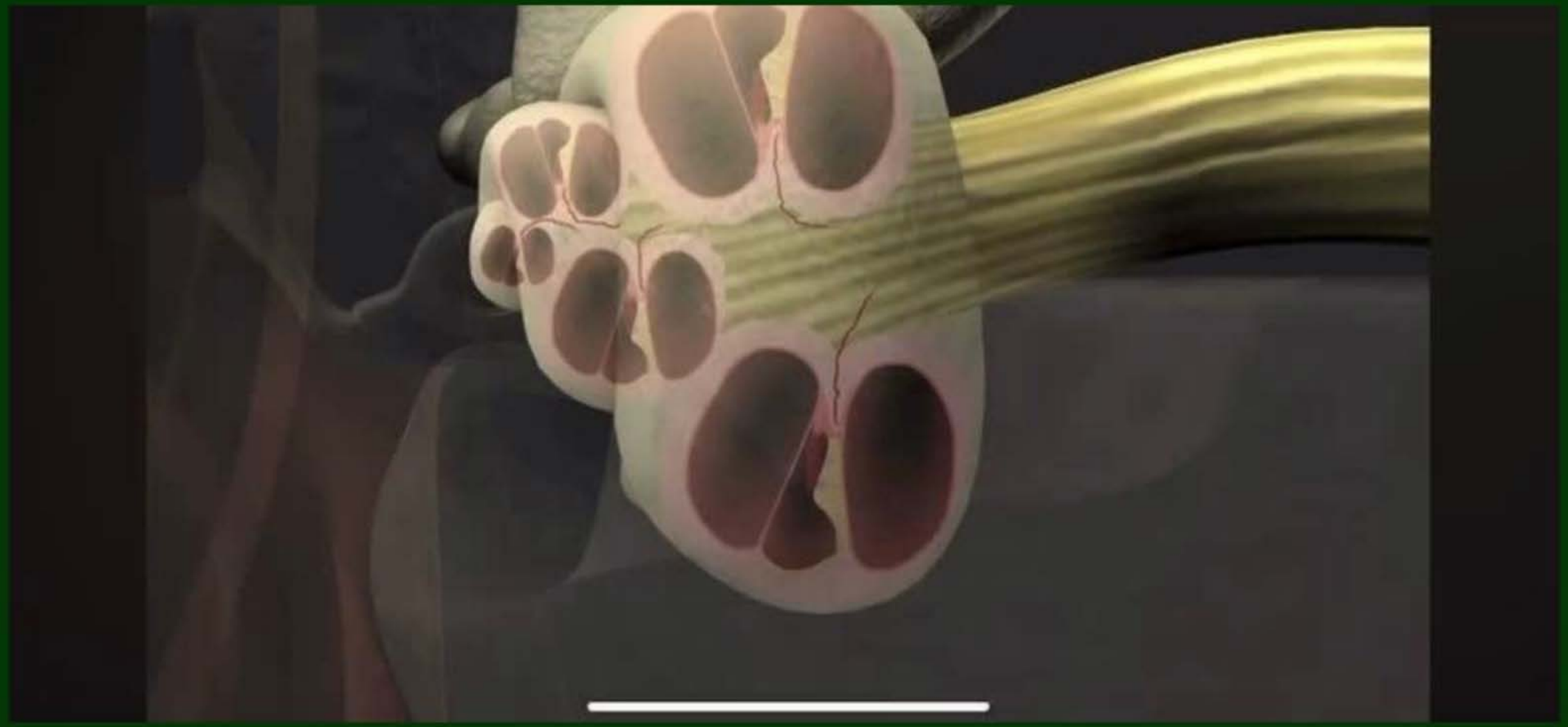
(d) What is the significance of the tiny hair present in the auditory canal? to trap germs. [Unde

(e) What is the function of auditory nerve? [Unde

aud.
to send impulse from Cochlea → brain

A) Auditory Nerve
B) Eustachian Tube


scala tympani
scala vestibuli
scala media



Class 10th ICSE Paper Discussion

(ix) Bharat woke up late in the morning and missed the school bus.



This situation stimulated the nerves of the sympathetic system which resulted in:

(a) Constriction of Coronary arteries

(b) Muscle relaxation

(c) Decrease in Respiration rate

(d) Bronchodilation

All arteries ↓ constrict BP ↑

O2

Class 10th ICSE Paper Discussion

Question 2

(i) Give the biological / technical terms for the following: [5]

(a) The tropic movement wherein the tendrils of a pea plant twine around a support.

(b) A defect in our eye in which some parts of the object are in focus while the other parts are blurred. Astigmatism.

(c) The type of waste generated in hospitals and pathological laboratories.

(d) The surgical technique for females that can be used to prevent pregnancy.

(e) The evolutionary process by which new species arise from the existing ones.

SECTION- B

Class 10th ICSE Paper Discussion

(ii) Given below is the picture of an eagle. [2]



depth / distance.
focus
3D image

Eagles have binocular vision. What is the advantage of such a vision?

Class 10th ICSE Paper Discussion

(ii) How are the Cytons and Axons of neurons arranged in the following? [2]

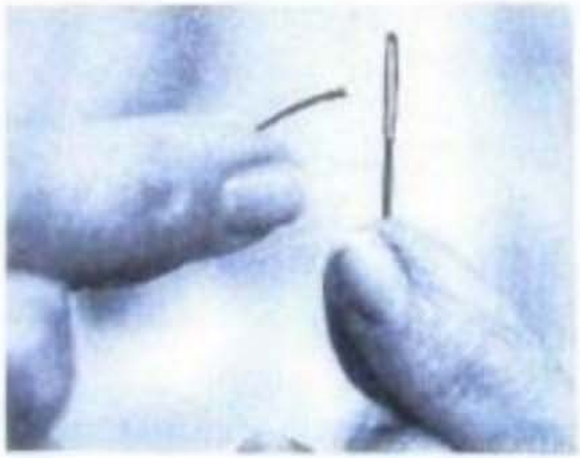
(a) Cerebrum

(b) Spinal Cord

H.W

Class 10th ICSE Paper Discussion

(v) Tara's grandmother is 70 years old and has a passion for embroidery. She faces difficulty in threading the needle as the eye of the needle appears blurred. The ophthalmologist diagnosed it as an age-related disorder. [3]



(a) Name the eye disorder she is suffering from.

(b) How can the above defect be corrected?

(c) Where is the image formed in the above disorder?

Presbyopia

behind the Retina

Thank You